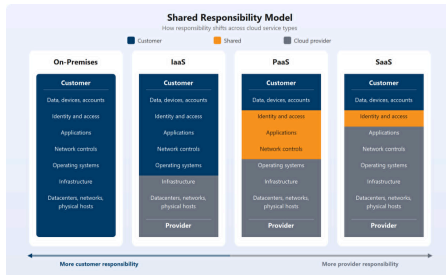


Cloud Computing - use of compute, network, and storage resources over the Internet under a consumption-based model

On-site Server, Public, Private, Hybrid, and Multiple Cloud

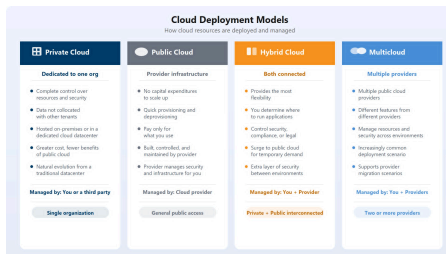


IaaS - lift and shift migration, testing and deployment use cases

PaaS - development framework, analytics and business intelligence use cases

SaaS - messaging, productivity, finance/expense tracking use cases

Shared Responsibility Model



Always the responsibility of the company

1. Information/data
2. Devices (mobile/PC)
3. Accounts and identities

Always the responsibility of cloud provider

1. Physical hosts
2. Physical network
3. Physical data center

Varies by type

1. Operating System - onsite and IaaS are provided by company
2. Network Controls

3. Application
4. Identity and directory infrastructure

Azure Account, Azure Free Account, and Azure Student Account

Azure Web Portal - Azure Website

Service Level Agreement (SLA) - Formal agreement between service provider and customer, Azure SLAs are represented as a **percentage (%)** of uptime, 99% uptime has a down time of 1.68hrs a week or 7.2hrs per month

Azure Arc - unified management view of resources from private cloud. Azure management and governance capabilities to on premises environments, like Kubernetes clusters and databases outside Azure

Azure VMWare Solution - VMWare workloads on Azure

Capital Expenditure (CapEx) - One-time upfront costs

Operational Expenditure (OpEx) - spending money over time, Pay As You Go is a similar term

TEN CLOUD BENEFITS

1. **High Availability** - accessible during failures and updates
2. **Scalability** - increase/decrease resource capacity based on workload. Azure supports manual and automatic options

Vertical Scaling (up/down) - increasing or decreasing resource capabilities, upgrading VM sizes like virtual CPUs / ram

Horizontal Scaling (in/out) - adding or removing more resources (instances), more vm's/containers. Azure Services like VM Scale Sets, App Service Plan Scaling, and AKS (Kubernetes) support horizontal scaling

3. **Elasticity** - automatic scalability
4. **Reliability** - recover after system failures (decentralized structure)
5. **Predictability** - forecasting performance or costs, data consistency

Predict performance - predict resources through autoscaling, load balancing, and high availability

Predict cost - predict expenditure through real-time tracking, data analytics, and pricing calculator

6. Security - data encryption, identity and access management, threat monitoring

IaaS - maximum control over security

PaaS/SaaS - automatic patches
Built-in protections like DDoS Protection

7. Compliance - auditing and mitigation for flagging of non compliant resources as well as low level scheduled patches and maintenance

8. Governance - predefined templates for adherence to corporate/government standards

9. Manageability - manage and deploy resources, tracking, and alerts

10. Sustainability - economies on scale, reduces an organization's carbon footprint

How THE CLOUD manages resources

1. Auto-scale - based on need
2. Templates - repeatable deployments
3. Health monitor - replace failing resources
4. Alerts - real time metrics

How YOU manage the cloud

1. Web Portal
2. CLI
3. APIs
4. Powershell

AZURE GEOGRAPHY

Availability regions - geographical areas with data centers, 60 plus regions in over 140 countries

Microsoft Entra ID, Azure Traffic Manager, and Azure DNS are not region-locked

Availability zones - physically separated data centers within an Azure region, with independent power, cooling, and networking if one fails (**isolation boundary**). They are connected through fiber.

Each region has AT LEAST 3 availability zones, but NOT ALL REGIONS SUPPORT

AVAILABILITY ZONES

Azure services that support availability zones fall into three categories:

1. Zonal services: You pin the resource to a specific zone (for example, VMs, managed disks, IP addresses).
2. Zone-redundant services: The platform replicates automatically across zones (for example, zone-redundant storage, SQL Database).
3. Non-regional services: Services are always available from Azure

geographies and are resilient to zone-wide outages as well as region-wide outages.

Region pairs - 482km/300 miles of separation, automatic replication for some services, region recovery, and sequential update rollouts (e.g., US East and US West). If one dies, the other is a backup

Sovereign regions - isolate from main instance for compliance or legal purposes (US Washington and China North/East)

Datacenters - physical infrastructure, data racks with pcs

Azure Virtual Machines (IaaS) - allows of hosting apps with high availability and fault isolation

VM Scale sets - creation and management of a group of identical, load-balanced, increasing or decreasing in response to demand

VM Availability sets - staggered updates and varied power and network connectivity. Availability sets group VMs by **(1) Update domain**: VMs that can be rebooted together during planned maintenance. **(2) Fault domain**: VMs that share a potential power or network failure point.

Azure Load Balancer - distributing web traffic across VMs, improved performance

Azure SQL Database - automatic failover, data durability

A RESOURCE IS THE BASIC BUILDING BLOCK OF AZURE. ANYTHING YOU CREATE,

PROVISION, DEPLOY, ETC. IS A RESOURCE

Resource group - when you create a resource, you need to place it in a group. **Resource groups are groupings of resources. Every resource must belong to exactly one resource group. You can move some resources between groups, but a resource is only associated with one group at a time. Resource groups can't be nested, and they can't be renamed after creation, so choose a clear naming convention from the start.**

Actions you apply to a resource group affect all resources inside it currently and in the future.

Deleting a resource group deletes everything in it. Granting or denying access applies to all its resources.

Azure Management groups - level of scope above subscriptions, apply governance conditions. 10000 management groups can be supported in a single directory, with up to six levels of depth

Azure Subscription - unit of management, billing, and scale
Types of Subscription Boundaries
1. Billing - billing requirements
2. Access control - organizational structure

Azure Virtual Desktop - desktop/app virtualization service that runs on the cloud
- access anywhere
- centralized security with Microsoft Entra ID (previous called Azure Active Directory)
- with MFA and Rbac
- multi session use, keeps data in cloud

Azure Container Services

- Auto scales with demand
- Pay only for runtime
- Supports stateless or stateful

1. Azure Container Services

(PaaS) - run containers without managing services. You upload your containers, and the service runs them for you.

2. Azure Container Apps (PaaS) -

build container apps that also include built-in load balancing and scaling, so your design can adapt to changing demand.

3. Azure Kubernetes Services - many containers together, it is a container orchestration service

Azure Functions (PaaS)

- event-driven serverless code, no VM/container, only runs when triggered
- runs in response to HTTP requests, Timers, and Azure Service Messages

Azure AI Services - provides prebuilt capabilities for common AI scenarios, such as language, speech, vision, and document processing. It is useful when you want to add intelligent features through APIs instead of training your own model first.

Azure OpenAI Service is another AI option in Azure that supports generative AI scenarios, such as chat and content generation, with built-in security and governance controls.

Azure IoT Services help you connect, monitor, and manage devices.

1. Azure IoT Hub enables secure, bi-directional communication

between cloud services and IoT devices.

2. Azure IoT Central provides a simplified software as a service (SaaS) IoT platform for solution builders.

3. Azure IoT Edge extends cloud capabilities to edge devices so some workloads can run closer to where data is generated.

Azure App Services

TYPES

- 1. Web Apps** - .NET, Node.js, Java, Python, PHP
- 2. API Apps** - RESTful APIs with Swagger Support
- 3. WebJobs** - Run background tasks/scripts
- 4. Mobile Apps** - Backend (auth, db, push)

KEY FEATURES

1. Easy deployment
2. Auto scaling
3. Secure endpoints
4. Supports CI/CD
5. Share infrastructure

Azure Virtual Network (VNet) -

lets you create multiple isolated virtual networks and communication between Azure resources, extension of your on premises network

SIX FEATURES

- 1. Isolation and segmentation** - private IP ranges, VPN gateways for virtual subnets and network segmentation), DNS service
- 2. Internet communication** - public IP or Azure public load balancers
- 3. Internal Azure communication** - connect vms, web apps, databases using Virtual Network or service endpoints

4. On Premise Connectivity / Network Security Groups

- Point to Site VPN for single device
- Site to Site VPN for entire networks
- ExpressRoute for private and high bandwidth link
- VNET Peering - Two Virtual Networks Between Azure

Network security groups are Azure resources that can contain multiple inbound and outbound security rules. You can define these rules to allow or block traffic, based on factors such as source and destination IP address, port, and protocol.

Network virtual appliances are specialized VMs that can be compared to a hardened network appliance. A network virtual appliance carries out a particular network function, such as running a firewall or performing wide area network (WAN) optimization.

EXPRESSROUTE is a DEDICATED PRIVATE CONNECTION between on premises network and Microsoft Cloud services, provided through a third-party connectivity provider, no traffic goes over the Internet. Global connectivity with **ExpressRoute Global Reach**

FIVE CONNECTION TYPES

1. ExpressRoute Circuit - Azure ExpressRoute lets you extend your on-premises networks into the Microsoft cloud over a private connection, with the help of a connectivity provider.
2. Colocation at CloudExchange - Use a cross connection at shared location

3. P2P Ethernet - direct line with you and Microsoft
4. Any to any (IP VPN) - through WAN provider
5. ExpressRoute Direct - straight to Microsoft Backbone with High Bandwidth, 100gbps

ExpressRoute enables direct access to the following services in all regions:

Microsoft Office 365

Microsoft Dynamics 365

Azure compute services, such as Azure Virtual Machines

Azure cloud services, such as Azure Cosmos DB and Azure Storage

5. Traffic management - Custom route tables and Border Gateway Protocol support, filtering with Network Security Groups and Network Virtual Appliance for Firewalls

6. Advanced Networking - VNet Peering - connect Vnets privately across regions, user defined routes, and public and private endpoints

Azure VPN Gateway - special Azure resources to connect on premises networks to Azure or connect individual devices to Azure or connecting Azure Virtual Networks together. IT USES THE PUBLIC INTERNET

Policy Based VPNs - specific IP rules

Route Based VPNs - use IP routing, better for complex changing networks

High Availability Options for VPNs

1. Active/Standby (Default): 2 instances, 1 failover

2. Active/Active: Both VPN handle traffic

3. ExpressRoute failover: use VPN as backup if ExpressRoute fails

4. Zone-redundant Gateways - spread across multiple availability zones

Azure DNS - manage and resolve DNS using MS Azure, converts domain names into IP addresses

Private DNS Zones - custom domain names within private virtual networks

Alias Records - automatically update DNS records when linked Azure Resources

Azure Storage Account - unique namespace to store and access data in Azure via HTTP/HTTPS - storage account names must be 3-24 characters, lowercase letters and numbers only, unique

Azure Storage Account Types

1. Standard general purpose v2 - most scenarios, all data types and redundancy types

2. Premium block blobs - Azure blob storage, only LRS and ZRS - high transaction rates or use smaller objects, consistent low latency

3. Premium file shares - Azure Files, only LRS and ZRS, enterprise or high performance scale applications, for SMB (all OS)/NFS (Linux) file shares

4. Premium page blobs - Page blobs only, LRS only

Azure Storage Redundancy

1. Locally Redundant Storage (LRS) - 3 copies, 1 Azure

datacenter, at least **11 nines** of durability

2. Zone Redundant Storage (ZRS) - 3 copies, multiple availability zones, at least **12 nines**

3. Geo Redundant Storage (GRS) - 3 copies locally, 3 more in paired region, at least **16 nines**

4. RA-GRS - adds Read Access

5. GZRS - ZRS + GRS

6. RA-GZRS - Read Access to the secondary region

Azure Storage Service Types

1. Azure Blob - scalable storage for unstructured data

2. Azure Files - manage cloud file share using SMB (server message block) and NFS (network file system)

3. Azure Queues - asynchronous messaging queueing / store between app components

4. Azure Disks - block level storages as virtual hard disk/ssd for vm's

5. Azure Tables - NoSQL storage, petabytes of data

Benefits of Azure Storage

- Durable and highly available
- Secure
- Scalable
- Managed
- Accessible

Azure Blob Storage

Blob Tiers:

-Hot for frequent, daily use

-Cool for infrequent stored at least 30 days,

-Cold for rarely accessed 90 days,

-Archive for long term storage 180 days

Considerations for access tiers:

Hot, cool, and cold access tiers can be set at the account level. The

archive access tier isn't available at the account level.

Hot, cool, cold, and archive tiers can be set at the blob level, during or after upload.

Cool and cold tiers have lower storage costs but higher access costs and lower availability SLAs. Archive tier has the lowest storage cost and highest rehydration latency and access cost.

Azure Queue Storage - large number for asynchronous processing, millions of messages, up to 64kb

Azure Migrate - Unified migration platform from on-premises to the cloud

Azure Data Box - physical migration service that helps transfer large amounts of data (80tbs) in a quick, inexpensive, and reliable way. The secure data transfer is accelerated by shipping you a proprietary Data Box storage device that has a maximum usable storage capacity of 80 terabytes. The Data Box is transported to and from your datacenter via a regional carrier. Once the data from your import order is uploaded to Azure, the disks on the device are wiped clean in accordance with NIST 800-88r1 standards. For an export order, the disks are erased once the device reaches the Azure datacenter.

AzCopy - command line utility, With AzCopy, you can upload files, download files, copy files between storage accounts, and even ONE WAY synchronize files.

Azure Storage Explorer - standalone app that provides a graphical interface to manage files

and blobs in your Azure Storage Account. It works on Windows, macOS, and Linux operating systems and uses AzCopy on the backend to perform all of the file and blob management tasks.

Azure File Sync - a tool that lets you centralize your file shares in Azure Files and keep the flexibility, performance, and compatibility of a Windows file server.

BIDIRECTIONAL DATA SYNC and CLOUD TIERING (frequently accessed files are local, rare accessed in cloud)

With Azure File Sync, you can: Use any protocol that's available on Windows Server to access your data locally, including SMB, NFS, and FTPS.

Have as many caches as you need across the world.

Replace a failed local server by installing Azure File Sync on a new server in the same datacenter.

Configure cloud tiering so the most frequently accessed files are replicated locally, while infrequently accessed files are kept in the cloud until requested

MICROSOFT ENTRA ID USES

Microsoft Entra ID is Microsoft's cloud-based identity and access management service. It lets you sign in and access both Microsoft cloud applications and cloud applications that you develop

1. Authentication
2. single sign on (SSO)
3. Application management
4. Business to business (B2B)
5. Device management

MICROSOFT ENTRA DOMAIN SERVICES - Microsoft Entra Domain Services provides

managed domain services — domain join, group policy, LDAP, and Kerberos/NTLM authentication — without requiring you to deploy or maintain domain controllers in the cloud.

- joining domains and applying group rules
- good for legacy apps

Can I connect my on-premises AD with Microsoft Entra ID?

Without a connection, an on-premises Active Directory deployment and a cloud Microsoft Entra ID deployment require you to maintain two separate identity sets. Microsoft Entra Connect bridges that gap.

How does Microsoft Entra Domain Services work?

When you create a Microsoft Entra Domain Services managed domain, you define a unique namespace. This namespace is the domain name. Two Windows Server domain controllers are then deployed into your selected Azure region. This deployment of DCs is known as a replica set.

You don't need to manage, configure, or update these DCs. The Azure platform handles the DCs as part of the managed domain, including backups and encryption at rest using Azure Disk Encryption.

Is information synchronized?

A managed domain is configured to perform a one-way synchronization from Microsoft Entra ID to Microsoft Entra Domain Services. You can create resources directly in the managed domain, but they aren't synchronized back to Microsoft Entra ID. In a hybrid environment

with an on-premises AD DS environment, Microsoft Entra Connect synchronizes identity information with Microsoft Entra ID, which is then synchronized to the managed domain.

Azure Authentication Method

1. **Passwords** - username and password
2. **SSO** - One login give access to multiple apps, most confident
3. **MFA** - 2nd layer , high security with reduced convenience (something you know, something you are, something you own)
4. **Passwordless** - no password but instead uses biometrics pins and secure keys

Azure Passwordless Options

1. Windows Hello - fingerprint/face login
2. MS Authenticator - Tap notification on phone
3. FIDO2 Security Keys - USB/Bluetooth/NFC hardware keys

Microsoft Entra External

Identities - people outside org, can use Bring Your Own Identity (BYOI)

B2B Collaboration - external users with own ID to sign in

B2B Direct connect - bi-directional trust with Entra ID

Azure AD (Entra ID) Business to Consumer or B2C

Azure Conditional Access and Role-Based Access Control

Microsoft Defender for Cloud

Defender for Cloud fills three vital needs as you manage the security of your resources and workloads in the cloud and on-premises:

-Continuously assess – Know your security posture. Identify and track vulnerabilities.

-Secure – Harden resources and services with Microsoft cloud security benchmark (MCSB).

-Defend – Detect and resolve threats to resources, workloads, and services.

Defense in Depth Model

1. Physical Security - datacenter hardware
2. Identity and Access - SSO / MFA
3. Perimeter - firewalls and DDoS protection
4. Network - segments resources and redundancy
5. Compute - secures vms, patches, endpoint protection
6. Application - build secure apps
7. Data - PII

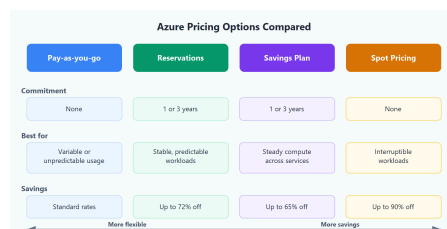
AZURE COSTING:

Many factors affect how much you pay. Some of the factors that affect cost are:

- Resource type
- Consumption
- Maintenance
- Geography
- Subscription type

Azure Marketplace

- Certified Azure applications and service with 10000+ listings



Azure Pricing Calculator -

Calculate cost of provisioned Azure Services, for future deployments
Basic configuration options include:

- Region
- Tier
- Billing options
- Support options
- Programs and offers
- Azure dev/test pricing

Azure Total Operations Cost

Calculator - Compare on-premises expenses to Azure cloud deployment

Microsoft Cost Management Tool

- helps monitor, control, and optimize Azure spending

Cost alerts

Cost alerts provide a single location to quickly check on all of the different alert types that may show up in the Cost Management service. **The three types of alerts that may show up are:**

-Budget alerts

-Credit alerts

-Department spending quota alerts.

Microsoft Purview - unified data governance platform, risk and compliance platform, provided visibility across your data landscapes, key capabilities include automated data discovery, sensitive data classification, and end to end data lineage

Azure Policy - DEFINITIONS OF THE CONDITIONS YOU WANT TO

CONTROL/GOVERN governance tool that helps enforce and audit resource compliance with organizational standards, restrict VM sizes and require tags

6 Effects in Azure Policy

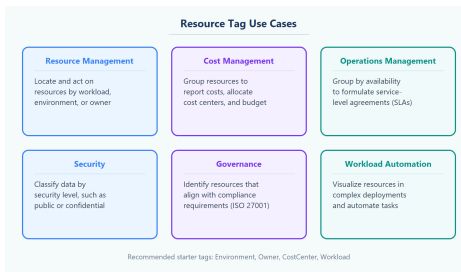
1. Append
2. Audit
3. AuditIfNotExists

4. DeployIfNotExists
5. Deny
6. Disabled

Azure Initiative - COLLECTION OF AZURE POLICY CONDITIONS

Azure Blueprint - SET OF STANDARDS, PATTERNS, AND REQUIREMENTS, for implementation of Azure cloud services, security and design

Resource Tags - provide metadata to Azure Resources, logically organizing into a taxonomy. A resource tag consists of a name and a value. You can assign one or more tags to each Azure resource.



Resource Locks - president accidental deletion and modifications, even by used with full access, adds extra layer

- **DoNotDelete Lock:** locks deletion, allows read and modify
- **Read only Lock:** blocks modification and deletion, allows read only access

Service Trust Portal - provides access to Microsoft's Security, Privacy, and Compliance resources.

Azure Powershell - POWERSHELL CMDLETS and REST APIS

Azure CLI - BASH STYLE SYNTAX

Azure Resource Manager (ARM) - deployment and management service for Azure, ALL ACTIONS

ON AZURE RESOURCES, via portal, CLI, SDKs, or APIs, go through ARM. **Templates are built and stored using JSON and have parameters. Declarative templates like JSON and BICEP, Infrastructure as Code**

Azure Key Vault is a service for securely storing and controlling access to:

- Secrets (such as connection strings and passwords)
- Encryption keys
- Certificates

Using Key Vault helps centralize secret and key management instead of storing sensitive values directly in application code or configuration files.

Azure Advisor - personalized recommendations Azure Advisor evaluates your Azure resources and makes recommendations to help you improve reliability, security, performance, and cost efficiency. Think of it as a personalized best-practices guide built into the Azure portal. Each recommendation includes a suggested action you can take right away, postpone, or dismiss.

You can also set up notifications so Advisor alerts you when new recommendations appear.

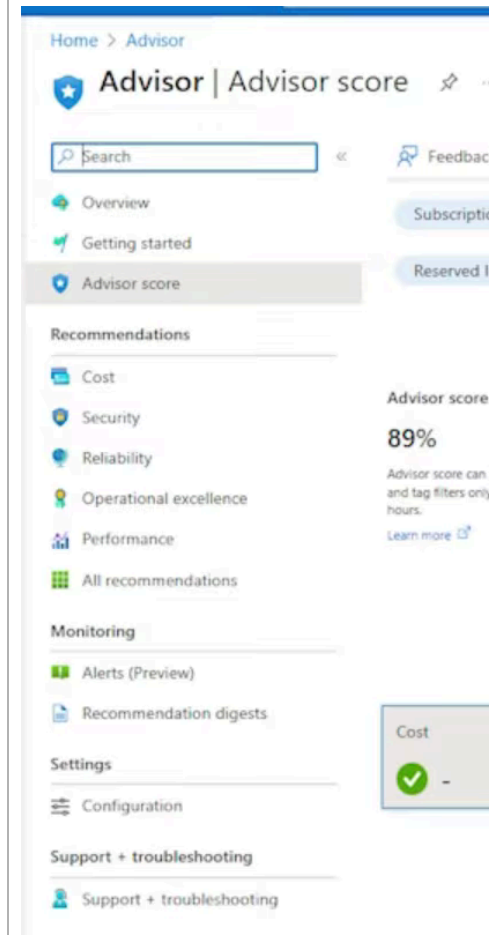
The Advisor dashboard displays recommendations for all your subscriptions, and you can filter by subscription, resource group, or service. Recommendations fall into five categories:

Reliability helps keep your applications running by flagging configuration risks.

Security detects threats and vulnerabilities that could lead to breaches.

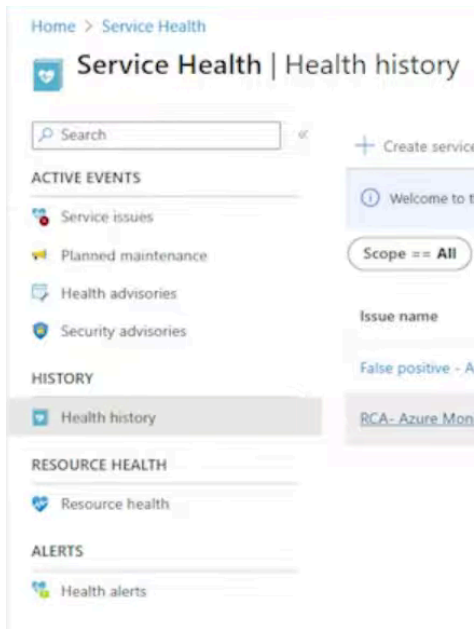
Performance identifies changes that can speed up your applications. Operational Excellence suggests workflow and deployment improvements.

Cost finds ways to reduce your Azure spending.



Azure Service Health - monitor the health of both the global Azure platform and your specific cloud resources

1. **Azure Status** - service outages
2. **Service Health** - personalized view of services/regions you use
3. **Resource health** - status of individual resources



Azure Monitor - records telemetry data from apps to get metrics and logs from all layers of network, you can analyze and visualize them using dashboards and Kusto queries, supports real-time alerts, and auto-scaling

Azure Log Analytics

Log Analytics is the tool in the Azure portal where you write and run queries against the data Azure Monitor collects. You can do simple filtering, like finding all errors in the last hour, or run advanced analytics to visualize trends over time.

Azure Monitor Alerts

Alerts notify you when Azure Monitor detects that a condition you defined has been met. You create an alert rule that specifies the condition, and an action group that controls who gets notified and what happens next.

Application Insights

Application Insights is an Azure Monitor feature that monitors the performance and usage of your web applications, whether they run

in Azure, on-premises, or in another cloud.

You can set up Application Insights by adding an SDK to your application code or by enabling the Application Insights agent without code changes.

Application Insights can monitor:

- Request rates, response times, and failure rates
- Dependency calls and their performance
- Page load times, user counts, and session trends
- Server performance counters such as CPU, memory, and network usage
- You can also set up availability tests that send synthetic requests to your application so you know it's responding, even during low-traffic periods.

Azure Cost Optimization Reservations

Reservations are best for stable, predictable workloads. You commit to specific resource capacity for a one-year or three-year term, and Azure applies discounted pricing to matching usage.

Azure savings plan for compute Azure savings plan for compute is another commitment-based option for compute services. Instead of committing to a specific VM family or instance type, you commit to an hourly spend amount for one or three years, and savings are applied to eligible compute usage.

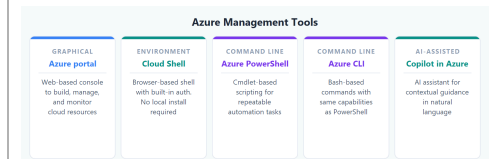
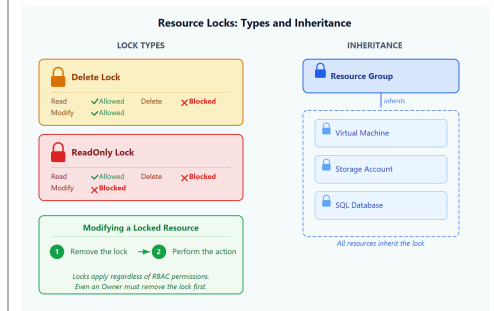
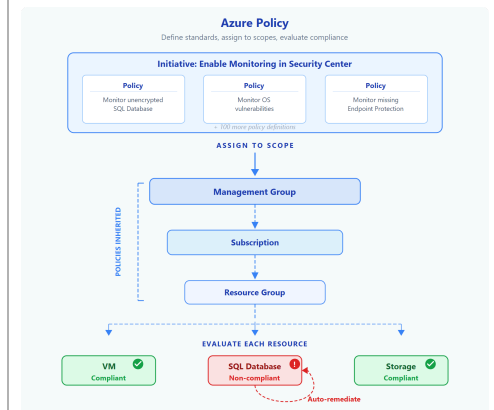
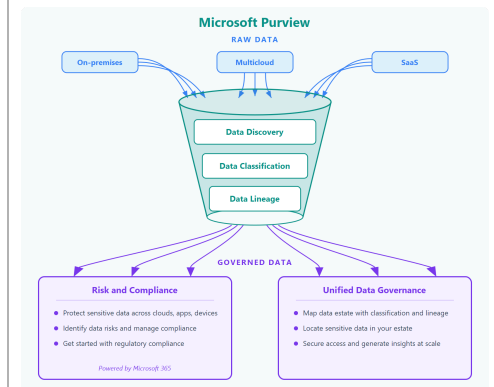
Spot pricing

Spot Virtual Machines use unused Azure capacity at reduced prices.

Spot is most appropriate for interruptible workloads because Azure can reclaim that capacity when needed.

You still get charged when you stop to close a virtual machine

ARM template - JSON file creates infrastructure quickly Azure Active Directory provides Authentication and Device Management



Azure Cloud Shell

Azure Cloud Shell is a browser-based shell tool that allows you to create, configure, and manage Azure resources using a shell. Azure Cloud Shell supports both Azure PowerShell and the Azure Command Line Interface (CLI), which is a Bash shell. **IT IS FOUND IN THE AZURE MOBILE APP AND ALL OTHER OS**

What is Azure PowerShell?

Azure PowerShell is a shell with which developers, DevOps, and IT professionals can run commands called command-lets (cmdlets). These commands call the Azure REST API to perform management tasks in Azure. Cmdlets can be run independently to handle one-off changes, or they may be combined to help orchestrate complex actions such as:

In addition to being available via Azure Cloud Shell, **you can install and configure Azure PowerShell on Windows, Linux, and Mac platforms.**

What is the Azure CLI?

The Azure CLI is functionally equivalent to Azure PowerShell, with the primary difference being the syntax of commands. **While Azure PowerShell uses PowerShell commands, the Azure CLI uses Bash commands.**

The Azure CLI provides the same benefits of handling discrete tasks or orchestrating complex operations through code. It's also installable on Windows, Linux, and Mac platforms, as well as through Azure Cloud Shell.

pwede kayo magreview para sa az-900 exam in 6-12hrs lang, review nalang kayo the day before para fresh sa utak yung info

[3hrs] Microsoft Learn AZ-900

Youtube Playlist -

https://youtube.com/playlist?list=PLahhVEj9XNTd6H2vpsvm_kBSMsMw9EMUL&si=SctcHL_XVloxWpOC

[2hrs 30m] AZ-900 Azure

Fundamentals Exam Cram - Full Course (Good in 2026!) -

 AZ-900 Azure Fundamentals ...

[40min] AZ-900 Exam Cram | Full Azure Fundamentals Course in 40 Minutes | Microsoft Certification 2025

 AZ-900 Exam Cram | Full Azu...

MS Learn Reviewer Course for AZ-900 [Course AZ-900T00-A] -

[Course AZ-900T00-A: Introduction to Cloud Infrastructure - Training | Microsoft Learn](#)